

Model migrations: mobility and boundary crossings in regional climate prediction

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The Hadley Centre's PRECIS regional climate modelling system has been designed to fulfil the informational requirements of adaptation and development planners in the 'global south'. Drawing on recent insights from science and technology studies and the geography of science concerning the mobility of scientific knowledge, this study investigates the institutional and discursive associations that enable the PRECIS modelling system to move between its UK birthplace and new sites of climate simulation. Document analysis and interviews with key personnel reveal the construction of regional climate modelling as an obligatory passage point for those seeking to adapt to future climates in developing countries. Furthermore, the operation of PRECIS across the boundaries of intersecting scientific and political worlds imbues the model with a level of epistemic power that has enabled the partial re-shaping of the global geographies of climate knowledge production. This new structuring of scientific practice is potentially empowering through the redistribution of climate modelling expertise, yet it may also contribute to the construction of climate prediction as a limit to adaptation. We argue that it furthers an epistemic hegemony that renders alternative 'ways of knowing' the climate either subordinate to or dependent upon the epistemic community centred on the Intergovernmental Panel on Climate Change and global governance mechanisms. The study illuminates the potential for geographers of science to make normative interventions in debates around the interplay of space, knowledge and power in contexts of environmental deliberation and governance.

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Introduction

Climate models are ubiquitous in discussions about the anthropogenic modification of the atmosphere. Modelling has emerged as a 'fundamental organizing principle for the global epistemic community that surrounds the climate change issue' (Sundberg 2007, 473). Processes of mutual exchange and reinforcement constitute a 'powerful and resilient set of relationships' (Shackley *et al.* 1998, 188) between modellers and policymakers, cementing the epistemic hegemony of simulation. They have also contributed to the construction of a particular form of political consensus around climate change (Demeritt 2001).

The PRECIS¹ Regional Climate Model (RCM) has been developed by the UK Met Office Hadley Centre (MOHC) with the aim of improving the information available to adaptation planners in developing countries through the generation of scenarios of possible future climate change on a regional scale. The system features a software package enabling the processing and display of data on any personal computer, and is provided to scientists in developing countries through partnerships with a number of governmental bodies at the UK and international level (Jones *et al.* 2004). The project began with a shared perspective between MOHC and the United Nations Development Programme (UNDP) on the need for scientific tools to aid

countries in pursuing adaptation policies, and more specifically in fulfilling their obligations to the UN Framework Convention on Climate Change (UNFCCC). The model is currently being used at meteorological agencies and scientific institutions in over 100 countries, and results have appeared in numerous policy reports, UNFCCC documents and scientific publications.

PRECIS is an example of a growing number of regional models being used internationally to add detail to the global projections produced by the major climate modelling centres (Rummukainen 2010). Substantial debate exists concerning best practices for the handling of uncertainty in regional modelling (Foley 2010) and the role that detailed climate scenarios should play in decisionmaking processes (Dessai *et al.* 2009). Questions about the political role of RCMs are more consequential in the case of PRECIS because of the model's unique associations with governmental structures and the political discourses of climate adaptation.

Using the tools of actor-network theory (ANT) (Latour 2005) and the social worlds framework (Clarke and Star 2008), this study explores how PRECIS has attained global mobility through the enrolment of a diverse set of actors and institutions within a network, and through its operation at the boundaries of various social worlds. The focus of the article is on the discursive construction of PRECIS as a scientific tool and of regional climate as a necessary object of study. The study seeks to contribute to the emerging literature on the geography of science, and to the growing engagement of social scientists with debates over the practices of climate science and computer simulation in policy-relevant contexts (Grundmann and Stehr 2010; Yearley 2009).

The following section provides an introduction to the theoretical perspectives employed in the study and a brief overview of the current literature on modelling practices. The research methodology is then described, followed by the main discussion and conclusions.

Geographies of science

There has been a growing focus within science and technology studies (STS) on the role of space and place in shaping the practice and influence of science across political and cultural contexts (Anderson and Adams 2008; Shapin 1998). Such questions do not just revolve around the significance of locality

in generating a universalising corpus of scientific knowledge (Rouse 1987), but instead address the 'distributed and relational' spaces of science (Livingstone 2005, 100). Increasingly scholars within STS, along with the new 'geographers of science' (e.g. Finnegan 2008; Hulme 2008; Livingstone 2003), have become interested in the *mobility* of scientific knowledge – questions of how scientific tools are able to overcome the friction of distance and attain 'usefulness' in new places, and the effects of these transfers on the epistemic landscapes of their new environments.

The question of how knowledge travels has been approached from a variety of disciplinary and theoretical perspectives, including the history of science (Secord 2004), material semiotics (Latour 1987), literary theory (Said 1983) and the nascent geographies of science (Meusburger *et al.* 2010). Each of these perspectives brings its own conceptions of spatiality to accounts of the mobility of knowledge (Powell 2007). This 'spatial turn' (Finnegan 2008) owes much to early work that emphasised the historical and material contingency of scientific practice (Knorr-Cetina and Mulkay 1983; Latour and Woolgar 1979). Such localist insights have generated questions regarding the transference of scientific knowledge and tools from context-to-context (Shapin 1995), while social scientists have become increasingly interested in the relationships between scientific and political cultures, and the performances of power across such boundaries (Jasanoff 2004). It is at this confluence of concerns for spatiality, knowledge and power in scientific practice that geographers of science have gathered (e.g. Hulme 2010).

The significance of space and place in shaping the discourses and practices of climate science has been explored by, for example, Hulme (2008), Jasanoff (2010), Lahsen (2004 2005) and Shackley (2001). In the context of atmospheric modelling, Jankovic (2004) has employed an explicitly geographical perspective in his account of how the Eta family of numerical weather prediction models, produced in Belgrade since the 1970s, have been able to travel between sites of knowledge production. Jankovic considers the contingencies – social, political, scientific, cultural – that make up assemblages of 'practice, instrumentation, theories and people' (2004, 48), to be key in understanding this mobility. The relative simplicity of the Eta models – a consequence of resource constraints – meant they were employed in the service of 'a

brotherhood of localities with similar conditions of scientific and social life' (2004, 64). His account highlights the significance of the dynamic and contingent 'micro/macro dialectics in a geographical distribution of knowledge', which shape the form and content of mobile tools and practices.

Networks, power, social worlds

As one of STS's most significant theoretical innovations, ANT has provided a theoretical framework and methodology for the study of the associations between human and nonhuman 'actors' across techno-scientific networks. ANT accounts for the processes by which actors enrol others into networks through the translation and alignment of others' interests with those of the researcher or technologist, thus enabling the extension and stabilisation of networks in space and time (Callon 1986). According to ANT, mobility demands that a complex assemblage of actors, ideas and tools behaves as a functioning network in order for scientific knowledge to travel in the form of 'immutable mobiles' (Law 1986) and thereby lay claim to the status of truth outside the laboratory (Latour 1999).

Overcoming the semiotic formalism of ANT – which tends to emphasise the structure of networks over the political relations they engender (Anderson and Adams 2008; Law and Hassard 1999) – may be aided by employing the *social worlds* framework. This is a symbolic-interactionist approach to social inquiry that draws attention to the interplay of power, knowledge and discourse across mutually-constitutive cultures. A 'social world' can be understood as an assemblage of people and institutions united in a shared discursive space (Strauss 1978), and a particularly salient line of inquiry has focused on the work that occurs at the boundaries of these spaces. The concept of the *boundary object* (Star and Griesemer 1989) has been used to explore how certain ideas, objects or practices enable exchange to take place between social worlds (e.g. science and politics) in a manner that accommodates both theoretical coherency and local plasticity (Star 2010). The meeting of diverse worlds often occurs within an *arena*, containing 'multiple worlds organized ecologically around issues of mutual concern and commitment to action' (Clarke and Star 2008, 113). Sundberg (2007) depicts 'climate change' as an arena in which a wide collective of entities interact (e.g. modellers, funding bodies, celebrities, activists). This interaction is enabled in part through the

operation of boundary objects such as the construct of global temperature, model parameterisations or even climate models themselves, all of which may be adapted for use by actors with diverse interests (e.g. Shackley 2001).

Close attention has been paid to the indeterminacies and contingencies of climate modelling practices (e.g. Heymann 2010), including their operation at the boundaries of various worlds or sub-worlds (Hulme and Dessai 2008; Shackley and Wynne 1996) and their role in the politics of adaptation (Dessai *et al.* 2009). But this critical gaze has not yet been fully directed towards the practices and technologies of regional climate modelling. Through studying the mobility of the PRECIS system, consideration is given to the translation processes by which actors have been enrolled into the PRECIS actor-network, to the work performed by the model at the boundaries of scientific and political worlds, and to the complex dialectical play of 'local' and 'global' in climate knowledge-making. It will be argued that analysis of this kind can sanction normative interventions in debates on the role of scientific knowledge in contemporary societies, a function that has largely been absent in recent work on the geographies of science (Powell 2007).

Methodology

The research took the form of a single case-study design. The bulk of the empirical data came from the analysis of key documents related to PRECIS including instruction manuals, promotional materials, scientific reports and presentation slides produced by MOHC staff. The majority of the 47 documents analysed are publicly available. Six semi-structured interviews were also conducted, with interviewees selected by theoretical sampling (Clarke 2005) and questions designed with the intention of exploring the different 'perspectives', i.e. the organised sets of ideas and beliefs orienting action and interaction (Becker *et al.* 1961, 34), which have coalesced around the PRECIS project. The interviews were thus broadly organised around themes of institutional relationships, the scientific contribution of PRECIS and its relationship to policymaking. Data collection also included a visit to the Hadley Centre's headquarters in Exeter and an informal group discussion with the PRECIS project team. Drawing on elements of discourse analysis and grounded theory (Charmaz 2006; Clarke 2005),

analysis proceeded through qualitative coding with constant iteration between the codes and concepts emerging from the data and the theoretical insights presented above. All interview extracts are presented with the permission of their sources.

A new geography of regional climate modelling

The spread of the PRECIS regional modelling system since 2002 has engendered a new geography of climate modelling practice. In 2001, only around 20 countries were running RCMs to the level of producing published research, the majority of which were OECD states (cf. Edwards 2010, 171). However, there are now 104 countries with researchers trained in the use of the PRECIS RCM. Due to computational demands, the form of these new communities of practice is often one of regional cooperation organised around nodal institutions or state offices. In addition, an international community of users has developed, with communication facilitated by an online discussion forum and a PRECIS 'group' on the social-networking website Facebook.² This is a novel situation, rooted in the MOHC's desire to create a tool with near-universal applicability. A collaborative relationship with the UNDP, which supports countries in preparing vulnerability assessments for National Communications to the UNFCCC, is testament to the model's broad applicability. Describing the evolution of modelling work through the cycle of preparing National Communications, a senior programme manager within UNDP states:

whether we think it's correct or not, the fact is that I will say the majority of the countries have a tendency to say now 'we have already done this work in the previous National Communication' (...) You can already see a lot of countries saying 'now we want to move to another phase of developing our scenarios that better reflects national circumstances' (...) and PRECIS is probably one of the tools that is most readily available to help them do so. (Interview 1 July 2010)

Here it is suggested that the form and rhythms of the National Communications process has influenced the content of the presented knowledge. The cycle of reporting, and the desire for statements to be updated in the light of advancing knowledge and policies, has influenced countries' desires to actively advance their knowledge through the seeking of more advanced tools. The establishment of

PRECIS as a desirable tool of climate prediction is the subject of the following section.

Making space for PRECIS

The PRECIS system was developed ... to provide a tool for predictions. It is made widely available to developing countries as an adaptation tool. (MOHC 2008, 1)

Translation

The dissemination of the PRECIS modelling system can be understood as an example of network-building – the construction of collective, heterogeneous relations that enable scientific facts and machines to find relevance and applicability outside of the laboratory or their equivalent place of production (Latour 1987). Key to the building of these networks are the processes of *translation*, which can be understood as the acts of persuasion by which an actor gains 'authority to speak or act on behalf of another' (Callon and Latour 1981, 279). Callon (1986) identifies four 'moments' of translation that enable actors in a technoscientific network to establish relationships and extend the network. The first of these, *problematization*, concerns how researchers 'seek to become indispensable to other actors by defining the nature and the problems of the latter' (Sundberg 2005, 37).³ Through the construction of 'funnels of interest', the general interests of other actors are channelled through the problem-formation of the researcher, 'in such a way as to establish themselves as an obligatory passage point in the network of relationships' (Callon 1986, 204). In order to extend the PRECIS network, the MOHC had to convince a series of other actors (UNDP, the UK Department for International Development (DFID), prospective users etc.) of the benefits of allying themselves with the project. Key to this was the construction of PRECIS as a useful adaptation tool.

In much of the material produced to accompany PRECIS, emphasis is placed on the essentiality of high-resolution regional climate scenarios for the development of adaptation policies. For example, in an early PRECIS brochure the first few paragraphs define the problem in question, and formulate a problem-solution:

All parties [to the UNFCCC] have a requirement under Articles 4.1 and 4.8, to assess their national vulnerability to climate change ... Assessment of vulnerability requires an estimate of the impacts of climate change,

which in turn is based on scenarios of future climate ... [which] are generally derived from predictions of climate change from global climate models ... [GCMs] may be adequate where the terrain is reasonably flat and uniform, and away from coasts. However, in areas where coasts and mountains have a significant effect on weather (and this will be true for most parts of the world), scenarios based on global models will fail to capture the local detail needed for impacts assessments ... The best method for adding this detail to global predictions is to use a regional climate model. (MOHC 2001, 1)

We can see in this extract that a funnel of interest is created, by which anyone concerned with fulfilling the UNFCCC requirements for vulnerability assessment must reach their goal through the practices of regional climate modelling. This and other documents have sought to establish regional modelling as an obligatory passage point for those seeking to understand their vulnerability to climate change. The MOHC's status as a 'pioneer' of this 'most proven method for obtaining detailed predictions' (MOHC 2001, 2) combines with the institutional authority of the project partners ('it was also useful for us [MOHC] to have the UNDP seal of approval on it' – former MOHC programme manager [hereafter 'MOHC PM'], interview 16 June 2010) to lend credence to the chain of translation. However, in some instances the problematisation has gone further, to present RCMs as obligatory passage points for the successful implementation of any kind of planned adaptation. In an MOHC document presenting an example of PRECIS's successful use in Bangladesh, it is argued that the vulnerability of Bangladesh's coastline makes adaptation a necessity, with potential activities channelled into the domain of the RCM:

Without this level of accurate, local information, countries simply cannot plan adequately for the likely impacts of climate change. (MOHC 2010, 1)

The point is reiterated later:

This is where access to high quality, high resolution climate information is so important. Put simply, credible adaptation policies cannot be created without such information. (2010, 5)

With these two statements RCMs and, by extension, PRECIS, are constructed as informational obligatory passage points for those actors seeking not only to fulfil the requirements of the UNFCCC, but also those charged with managing and avoiding the risks posed by climate change to the

infrastructures, livelihoods and cultures of contemporary societies in Bangladesh. With this example presented here as a 'model' for other developing countries to emulate, the translation of interests is extended across global space.

Vulnerability

With their limited resources, infrastructure and dependence on subsistence agriculture, developing countries are acutely vulnerable to the potential impacts of climate change – and none more so than Bangladesh. (MOHC 2010, 1)

The vulnerability of certain developing countries is often placed in the foreground of any account of the necessity of regional prediction. In the above, vulnerability (as a function of macro-economic variables at the national scale) is invoked to establish the need for more knowledge. Such references to vulnerability as pertaining to national economies vastly outnumber statements concerning the vulnerability of societies or communities (cf. Barnett and Campbell 2010). This can perhaps be seen as a strategy to couch vulnerability in the language of decisionmakers (e.g. economics) in order to convince them of the applicability of the RCM approach to their informational requirements. This sentiment is expressed by DFID – a key project partner and funder of training activities – in a briefing document aimed at DFID staff and 'development partners' concerning the necessity of scientific information for adaptive decisionmaking in developing countries. We hear of the

need for clearer messages concerning short and long term climate risks, including numbers of people affected and estimated economic impacts, that enable governments' planning and finance departments to act. (DFID 2004, 2)

With such emphasis on the quantification of possible climate impacts and the widespread use of an economic vocabulary in communicating results to policymakers, it is understandable that vulnerability is presented in the language of macroeconomics to interest decisionmakers in the tool. However, vulnerability in other cases is rendered as something environmental in nature, defined by the occurrence of climatic extremes and hazards. For example, the application of PRECIS in Bangladesh is linked to a measure of such vulnerability:

Why does Bangladesh need this project? The geographical location of Bangladesh makes it the most vulnerable

country in the world to tropical cyclones and their associated storm surges ... the warm waters of the Bay of Bengal generate intense tropical storms which often tear through the country with devastating impact. (MOHC 2010, 4)

In this extract, particularly emotive language ('tear/devastating') is employed to link the need for regional climate projections with a need to reduce societal vulnerabilities to extreme climatic events. It is of course undeniable that climate change could have very serious impacts for many communities in places such as Bangladesh. However, it is essential to contemplate how concepts and discourses such as 'vulnerability' are mobilised, in order to ensure that they serve the people they seek to represent in the best possible way. Bankoff (2001) reflects on how 'vulnerability' can be seen to be part of a genealogy of totalising, 'othering' discourses that construct a discursive geography of dangerous spaces and places delineated by climatic excess, natural hazard and societal frailty (Gaillard 2010; Manzo 2010). This in turn 'generates a moral obligation on behalf of Western nations' to render these regions safer, primarily through 'the transfer and application of Western expertise' (Bankoff 2001, 27).

Elements of this discursive reading of vulnerability are reflected in the deployment of the concept within the context of promoting PRECIS as a useful adaptation tool. Developing countries are portrayed as being distinctly or 'acutely' vulnerable to certain aspects of climate change, and 'credible' policies to reduce that vulnerability and adapt to changing circumstances can only be enacted with the aid of scientific tools developed in Western states. The PRECIS project can therefore be said to be participating in the construction of a discursive geography of danger and vulnerability. Although vulnerability as a concept can be usefully deployed by states or communities to gain access to international resources, this dominant discourse of vulnerability risks collapsing our imaginations of societal futures into a new form of environmental determinism, with climate as the dominant factor shaping the trajectory of our societies and cultures (Barnett and Campbell 2010; Hulme 2011).

Realism and the creation of reference

With vulnerability understood as being partly determined by local climate variability and extremes, potential users of PRECIS must be con-

vinced of the model's ability to effectively simulate their local atmospheres. References to how 'realistically' RCMs in general or PRECIS in particular simulate local climates are frequently made in MOHC documents, with statements such as 'RCMs simulate current climate more realistically' (Jones *et al.* 2004, 18) and accounts of local climatic features being 'skilfully resolved' (MOHC 2001, 14) being common. Terms such as 'detail/resolution/realistic/comprehensive' frequently occur in presentations made by MOHC staff. The PRECIS *Dissemination strategy* opens with an account of the potential of RCMs to offer high levels of realism, which is linked to the need to address vulnerability:

[RCMs] provide climate information with useful local detail including realistic extreme events. Developing countries are the most vulnerable to climate change. Hence the need for detailed climate change scenarios to assess their national vulnerability. (MOHC 2003a, 1)

In this formulation the provision of realistic simulations is obligated by the existence of high levels of vulnerability. However, as Latour (1999, 310) has argued, it is not sufficient to simply make such claims of realism without some instance of producing *reference* – 'the train of transformation' that links the empirical world to the resultant series of inscriptions produced by scientific practice, for example texts, diagrams, maps. It is the process by which 'data is transformed into representations of data and temporarily fixed' (Sundberg 2005, 38). Producing reference is therefore a means of demonstrating that the atmosphere that has been transformed through simulation and experiment can tell us something about the *real* atmosphere – a key means of generating credibility and legitimacy for modelling exercises.

In the case of PRECIS, the production of reference is slightly different to that in Sundberg's (2005) sociological account of the comparison of modelled and observed data. Credibility for PRECIS is sought through the frequent presentation of graphics highlighting the difference made by high resolutions to the representation of familiar geographic forms. In Figure 1a, a comparison is presented of the representation of the Philippines at various spatial resolutions (400, 50 and 25 km). As the resolution of the model increases, the shape of the coastlines becomes more detailed, more islands appear on the map, and the overall picture becomes one of topographical clarity with the

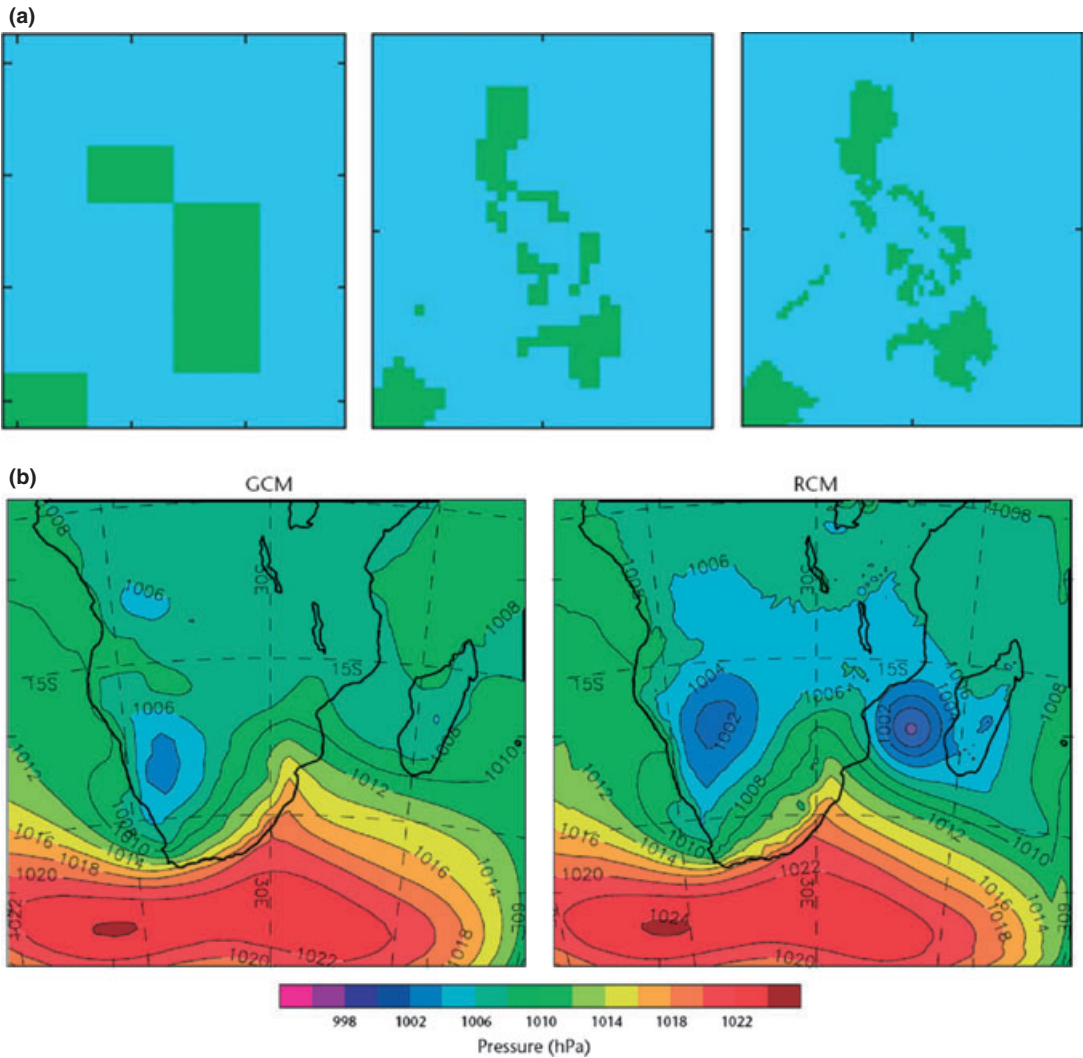


Figure 1 The production of reference

Sources: MOHC (2001); Jones *et al.* (2004). (©) Crown copyright 2001–2004, the Met Office

islands recognisable to anyone familiar with the geography of the western Pacific.

Figure 1b shows two representations of modelled air pressure patterns over southern Africa. Crucially the right-hand representation, drawn from an RCM, is marked by the appearance of a cyclone in the Mozambique Channel.

If we understand the positioning of these two groups of inscriptions in a number of MOHC documents as a production of reference with the aim to convince, we can begin to understand how ‘allies’

are recruited in support of modelling practices. However, unlike Sundberg’s (2005) example of confronting models with empirical data, in this case it is not two scientific representations that are being made ‘equivalent’. Rather, in Figure 1, the inscriptions of the highest resolution are appealing to knowledge of physical realities constructed through bricolage and experience, rather than scientific observation. In the case of Figure 1b, the recognition of the ‘existence’ of a cyclone in the model output brings this representation into greater

equivalence with extant understandings across social worlds of what is relevant to political decisionmaking, and of what is threatening and violent in the prospect of a changing climate. For Sundberg the creation of reference helped to establish legitimacy between the sub-worlds of climate simulation and observation, and within the wider social world of climate science. In this case, epistemic authority is sought for PRECIS across a range of social worlds, including those of climate policy and adaptation planning – in short, the broader *arena* of climate change.

Knowledge transference: from place-to-place and global/local

It is now a national demand to understand and predict the future climate. Hence, developing capacity to simulate, interpret and analyze future climate using climate models at a regional scale has become essential ... [PRECIS] is expected to be instrumental to enhance Bangladesh's capacity on climate change modelling. (BUET 2010)

In addition to PRECIS' role as an 'adaptation tool', one of the key purposes that the PRECIS project is seen to serve is the building of scientific capacity in developing countries, as evinced by the above quotation from a local report of a PRECIS workshop at the Bangladesh University of Engineering and Technology (BUET). This capacity-building is made up of two parts. Firstly, it implies technological capacity – PRECIS is given to countries as a tool that they can use, often alongside the provision of additional computational resources. Secondly, capacity-building involves the transference of the necessary knowledge and skills to be able to work with PRECIS in a way that the MOHC deems appropriate. The majority of this transference takes place at training workshops, where prospective users gather to be introduced to the system and to be offered the necessary guidance to continue working with it almost independently, albeit with *ad hoc* technical guidance.

Training, trust and expertise

Like most experimental work, replication cannot be guaranteed simply by written accounts and as such hands-on instruction is required to enable PRECIS users to generate scientifically credible results (Collins 1985). Hinting at early shortcomings, MOHC PM describes how the training programmes have

gradually become more comprehensive after it was found that in previous efforts some of the knowledge 'didn't really stick':

we evolved training courses which were much more comprehensive, so consisted of three or four people going out, spending a whole week, hands-on stuff, really going through things in detail. Lectures, hands-on experiences and so on. Getting the model to run, getting output, and that seems to have been successful.

In the sequence of documents produced to accompany the training programme, MOHC has placed progressively greater emphasis on the issue of uncertainty. Between two early versions of the PRECIS brochure (MOHC 2001 2002) and the production of the *Handbook* (Jones *et al.* 2004), 'uncertainty' went from being a topic under the 'Limitations of RCMs' in section 7, to being a stand-alone section in itself towards the beginning of the document – section 3. Despite this, many of our interlocutors have suggested that PRECIS has often been applied without a full treatment of uncertainty. MOHC trainers often bemoan the fact that they cannot get to 'the right people', i.e. that instead of training 'the person who was actually going to have the hands-on job of running the model', the trainers would be faced with 'his [sic] boss or his boss' boss' (MOHC PM, interview). Discussions of future workshops featured speculations on how to ensure the attendance of the modellers, rather than the managers, while promotional documents stress the importance of high-level knowledge of the climate system as a prerequisite for workshop attendance.

We can see through the example of the training process that the PRECIS project is not generating an entirely new community of practice through pedagogy, but is rather working within an existing, dispersed community of climate scientists, and promoting coalescence around a particular shared practice. This raises questions concerning the local plasticity of modelling practices (as opposed to a rigid standardisation), and the potential for practitioners to employ their own expertise in enhancing the 'skill' of the model for their own needs. Documents produced by MOHC and partner organisations often make use of terms such as 'local knowledge', implying that the modelling practices include space for local negotiation, modification and compromise. For example, a text summarising a meeting between DFID officials and MOHC scientists in 2003 outlines the perceived benefits that localised modelling will generate:

generating national predictions and scenarios in national centres of expertise gives the country ownership, builds capacity, and improves the predictions through input of local expertise. (DFID 2003, 3)

However, it is unclear what the form or content of this local knowledge or expertise should or can be. For MOHC PM, the incorporation of local knowledge could mean that users are able to adjust the model to better suit their own purposes. He states that there was an intention that:

people who were already reasonably well spun-up on modelling, like Cuba, particularly India and so on, would be able to dig-in to the model, the PRECIS model itself, and decide if some of the parameterizations or some of the things that were set in there could be done better over their particular parts of the world.

Subsequent discussions with members of the PRECIS team revealed that in certain cases this has happened. User groups who we have heard described as 'trusted collaborators', including groups in India and South Africa who had been involved with the initial development of the model, had begun to 'tweak' PRECIS (cf. Knorr-Cetina 1981) through adjustments to the MOHC's unified model. The ability of groups to do such work seems to be dependent not only on their modelling expertise, but also on the level of trust existing between them and the MOHC. This may well be due to the fact that any work subsequently carried out with the 'tweaked' model would still carry the Hadley Centre label, and as such would still be a vehicle for the Centre's international reputation. This demonstrates that trust between institutions and individual scientists is a key determinant of the strength of any scientific network in which knowledge production under a singular project or rationale is distributed across space. As Shapin argues, 'the recognition of trustworthy persons is a necessary component in building and maintaining systems of knowledge' (1995, 302).

Despite these few examples of users tweaking the model, it is acknowledged by MOHC staff that for most PRECIS remains a 'black box' (Callon and Latour 1981, 285), i.e. users can only take concern with the 'inputs' to the model (e.g. different boundary conditions, emissions scenarios, and the domain size and location) and its outputs (i.e. the regional scenarios). This is reflected in the users' licence agreement: 'The Licensee may not alter or modify the whole or any part of PRECIS' (MOHC 2003b, 2).

It is commonly argued that the trend for 'open source' simulation software/models (see <http://climatecode.org>; Merali 2010), which is often accompanied by a rhetoric of transparency and reflexivity, is in fact characterised by a high degree of *epistemic opacity* (Humphreys 2009, 5) which engenders an 'inability of those working with simulations to fathom the design principles and assumptions of the simulations and models' (Kouw 2010, 5). The inability of the majority of PRECIS users to 'dig into the model' reflects a degree of epistemic opacity, tempered only by the technical detail of the accompanying *Handbook* and the user's trust in the scientific credibility of the model constructors. For Kouw, epistemic opacity creates a new kind of vulnerability – 'risks in the form of blind spots, inscription [of unspoken values], and uncertainties' (2010, 1). The reliance on simulation to tell us about the operation of physical systems may expose us to the risk of overlooking parts of a system that science has not been able to codify in a model, while on the scale of an individual user, epistemic opacity may create a risk of misattribution of phenomena or a misunderstanding of what constitutes an epistemically relevant process to the attainment of particular practical or experimental goals (Humphreys 2009).

The rationale for PRECIS' opacity was to create a model that would be easy to use, and discussions with team members revealed that they are very proud of the model's ease-of-use and use it as a criterion by which to judge PRECIS against other RCMs. However, this level of user-friendliness is seemingly a product of negotiations made during the development process:

I know we talked about that a lot, and said 'well you don't want to make it too easy otherwise you'll get people who are using it who don't understand what they're doing, just clicking buttons and moving things around.' (MOHC PM, interview)

This statement reflects an awareness of the problems of epistemic opacity. Indeed, the issue of 'whether the user needs to know the details of the processes between input and output in order to know that what the instrument displays accurately represents a real entity' is still an unresolved question among philosophers of science (Humphreys 2009, 6). However, in order to extend the PRECIS network, the MOHC had to strike a balance between total reflexivity and ease-of-use. A 'traditionally' crafted RCM housed within the Met Office's supercomputers

wouldn't have been sufficiently mobile to find application in new situations. On the other hand, an entirely clement model demanding no prior expertise would quite likely be subjected to inappropriate use, damaging the scientific reputation of the MOHC. The result was a tool that the majority of users must accept as a black box.

Local knowledge/global praxis

Sentiments that local knowledge can improve regional prediction are characterised by ambiguity, particularly with regard to the relative epistemic opacity of the model. Statements such as 'it is vital these countries are able to make their own future predictions and combine them with local knowledge' (MOHC 2008, 1) provide few clues as to the meaning of 'local knowledge'. An academic researcher and consultant on climate change issues, who has worked with PRECIS in a number of internationally collaborative studies of climate change impacts in China, surmises thus:

I would guess that what they're [MOHC] talking about there is that they're interacting with local climate scientists and they are, through the process of running PRECIS and validating PRECIS they're getting information back from the community about how well the model is performing, simulating aspects of regional, local climate, and that helps them in terms of their subsequent design, re-design of PRECIS or even for Hadley global models and that that interaction is a two-way process, so it helps both sides. (Interview 25 May 2010)

Here it is suggested that the improvement of simulations occurs through improvements made to the model at the MOHC, rather than at the institutions running PRECIS. Through an iterative process, local expertise is incorporated into the model through the performance of scenario production itself. Put another way, local expertise does not constitute an initial 'input' to the model, but rather can be assimilated into the process through the running of the model, and comparison of the outputs to local expertise regarding regional climates. This suggests that 'local knowledge' may be shorthand for the knowledge embodied in local observational datasets. This is supported by statements in the PRECIS *Handbook*, e.g.:

It will also tap into valuable local knowledge, for example in checking the validity of models against observations from the region of interest. (Jones *et al.* 2004, 6)

Therefore 'it is desirable to have good observations of the climate relevant to the region' (2004, 21).

This need for 'good' local observational datasets has hampered some regional modelling activities where data inhomogeneity has made model validation difficult. This element of the socio-technical network of climate prediction is something that is largely external to the assemblage of relations and objects over which the MOHC is able to exercise control. That said, there is some evidence that PRECIS itself may have acted as a catalyst for greater observational efforts, through notionally linking together the capacity to observe with the ability to predict. For example, MOHC PM recounts how the PRECIS team became involved with the World Meteorological Organization (WMO):

WMO did a series of workshops on the GCOS⁴ programme in the different WMO regions and they asked us to talk about the regional climate model as a tool in its own right but also a reason why good observations should be made. So the WMO workshop was about observations mainly (...) And I gave this talk at this meeting, I think in Costa Rica, and they were very pleased with that, and we then got invited to subsequent, I think there were eight of them altogether in different parts of the world, so we were invited to talk at each of these and put the same message across.

This account suggests that the goals of the MOHC of using local observations to validate PRECIS and the goals of the WMO in generally encouraging observations were brought into alignment. With the full weight of the WMO and the MOHC behind the drive to promote observations as a means to achieving the ability to predict, this element of 'local knowledge' – for so long a domain where practice depended on historical contingency (Edwards 2010, 189) – is enrolled into the discursive and socio-technical networks of the climate change arena, largely in a legitimating role through the extension of reference between model simulations and their empirical grounding.

In addition to PRECIS' role in simultaneously harnessing and producing 'local' knowledge, it is maintained that results should contribute to the growing body of regional climate knowledge embodied by IPCC assessments. In the following communication to the UNFCCC, the Caribbean Community Climate Change Centre (CCCCC) expresses a wish to respond to a call from the IPCC for more knowledge regarding the Caribbean region:

The results will also prove valuable for the preparation of the IPCC Fifth Assessment Report. The IPCC noted

that there was less new scientific data from the Caribbean to prepare the Fourth Assessment Report than there was to prepare the Third Report. The Centre will ensure that this does not reoccur. (CCCCC 2008, 3)

The inclusion of PRECIS results in the IPCC reports has a distinctive geography itself, largely corresponding to the location of modelling groups who were involved in PRECIS' early development. As more groups advance in their use of PRECIS, the cartography of RCM research featured in IPCC reports is expected to change greatly. Thus it is clear that work with PRECIS is not simply about enhancing the knowledge of local scientists in order to bolster local decisions. Rather, the activities are intimately bound in a global praxis; an epistemic harmonisation by which the networks of climate science and politics, with their attendant practices, discourses and expectations, are extended and strengthened in global space.

Awareness

One of the many rationales linked to the PRECIS project is the capacity of scientific information concerning regional changes to generate increased levels of 'awareness' of the climate issue. This intention is reflected in the *Handbook*:

This project will lead to much more effective dissemination of scientific expertise and awareness of climate change impacts than could be achieved by simply handing out results generated from model runs in developed countries. (Jones *et al.* 2004, 6)

The effective spread of information and 'awareness' is presented as being dependent on the new geographies of knowledge production engendered by the PRECIS project. The assumption is that the production of scientific knowledge within the same political space in which it is to be employed enhances the ability of that knowledge to generate awareness. But this introduces the question of what is meant by 'awareness' as a form of knowledge itself. If understood in a discursive sense (cf. Foucault 1972), the mobility of PRECIS may contribute to the expansion of the 'discourse coalition' (Hajer 1995) centred on climate change. For Hajer, actors and institutions are shaped by their discourses and cognitive horizons, with specific ensembles 'of ideas, concepts and categories' (1995, 44) constraining action whilst simultaneously creating space for social change. PRECIS has the potential to introduce new ideas and concepts to political actors and institutions, which may

re-frame conceptions of the relationship between society and climate (Miller 2004) and generate the necessary conditions for political action. The intention of generating awareness can therefore be understood as an attempt to influence the dominant discourses of political and public institutions.

The task of raising awareness is often posited as being suited to the work of RCMs, with their operation on relatively small spatial scales helping to 're-localise' abstract global information to scales of personal and political interest (Jasanoff 2010). This type of argument is often couched in terms of making the issue 'visible', which reflects the concern with realism discussed above. In the case of PRECIS, users are advised to follow a successful example of this kind of work in the UK:

[It] can be valuable in its own right to publish this information ... in the form of maps and diagrams ... In the UK, we have found that, when combined with information on the current observed climate, a short booklet can be produced which is not only a source of information, but which can have the effect of making the issue of climate change more visible to government and other stakeholders. (MOHC 2001, 10)

This separation of 'information' and 'issue visibility' suggests that the mobility of PRECIS is predicated less on the existence of the necessary expertise and more on a widespread desire to influence political action. In the Caribbean, significant challenges exist in the operation of the model and the interrogation of uncertainties, largely attributable to a shortage of computational and human resources (Taylor *et al.* 2007). While this puts the burgeoning Caribbean modelling work into conflict with many of the norms developed in the field of regional climate modelling (e.g. Foley 2010), this has not precluded the use of model results to support political calls on the international stage for a 1.5°C cap on temperature rise (CCCCC employee, interview 2 July 2010). PRECIS is often presented as being able to facilitate this kind of influence through providing local 'ownership' of scientific information, despite the licence agreement suggesting that this ownership is not literal. While PRECIS thus has the potential to extend the discursive network of climate change to new political actors and institutions, it should also be understood as a mobile actor *within* a discourse coalition. PRECIS links concerned actors and institutions with scientific information and particular ideas on how best to predict and plan for climate change adaptation.

Conclusion

During the research we often encountered the metaphor of 'spinning-up' a modelling community. Before a model can be used to provide predictions, it must go through a 'spin-up phase' whereby the various elements are brought into dynamic equilibrium. The use of this metaphor for both the establishment and prior existence of expert knowledge communities emphasises the point that the MOHC is not simply engaging in what is normally delineated as 'scientific' work (Latour 1987), or even 'capacity-building'. Rather, through translating the interests of various actors through the production of a mobile tool of climate prediction, the Centre is engaged in re-shaping the materialities and social relations of this scientific practice.

The preceding discussion has demonstrated how the PRECIS regional modelling system operates at the interface of climate science, national and international climate policy deliberations and wider concerns regarding societal vulnerability to natural hazards. Through the extension of the PRECIS actor-network, new sites have been established where this complex intersection undergoes constant performance, renegotiation and representation. From a social worlds perspective, PRECIS can be seen to facilitate interaction and exchange between a number of worlds and sub-worlds: model developers, the climate impacts community, global and national political assemblages, non-governmental institutions etc. The climate arena, within which the various actors interact, provides a *transaction space* (Nowotny *et al.* 2001, 146) whereby asymmetrical relationships of dependency can develop at institutional and disciplinary boundaries. The exchanges that characterise transaction spaces often involve multiple translations, directed by powerful actors and facilitated by boundary objects that 'both inhabit several communities of practice and satisfy the informational requirements of each of them' (Bowker and Star 1999, 297). Callon argues that boundary objects are 'polysemous in the case of notions and statements, multifunctional in the case of technical devices, complex in the case of embodied skills' (1995, 59). PRECIS' multivalent purposes, as articulated by developers, partners and users, its notional flexibility engendered by its mobility, and its ability to fulfil a range of substantive, instrumental and discursive demands make it eligible for this description.

The production of PRECIS as a mobile boundary object facilitating interaction between scientific and

political worlds is intimately linked to the construction of RCMs as obligatory passage points for the attainment of a particular political sagacity. This is achieved through the translation of instrumental goals and the deployment of normative discourses of vulnerability and scientific realism, the consequence being a community pursuing knowledge that possesses high spatial resolution and precision. This pursuit is facilitated by the rendering of planned adaptation as captive to, or an ancillary of, the ability to predict future climatic changes on the scales that most interest decisionmakers. Prediction is therefore constructed as a *limit* to adaptation (Dessai *et al.* 2009). This privileges a certain approach to adaptation over others while potentially generating vulnerability through the pursuit of optimal, rather than robust, adaptation strategies (Dessai and Hulme 2007). Despite the construction of 'local' knowledge as both input to and output from the project, the epistemic landscape within which PRECIS moves is shaped by certain powerful actors, from the authoritative Hadley Centre to the ubiquitous UNFCCC. The approach to knowledge-making which PRECIS embodies is emblematic of the top-down approach to adaptation labelled by Dessai *et al.* (2005) as the 'IPCC approach' (see also Parry and Carter 1998), and the work of the IPCC is ever-present as a legitimating, custodial and aspirational force (see CCCCC quotation above).

The geographic expansion of science can be seen as part of wider processes of standardisation and rationalisation that characterise the condition of modernity (Jasanoff 2006). The harmonisation and mutual construction of scientific, bureaucratic and economic cultures (Jasanoff 2004) is evident in the interaction of the PRECIS system with the world of climate politics. Such a discursive hegemony renders ever more political decisions subject to the knowledge claims and political structures of the epistemic community that surrounds climate change, at the expense of alternative 'ways of knowing' (Hulme and Dessai 2008, 67). This hegemony brings questionable benefits. In the field of adaptation, we can observe a privileging of model predictions 'over explorations of how the many other dimensions of cultural, social and political life may change' (Hulme 2011, 262), thus privileging climate as 'the chief determinant of humanity's putative social futures' (2011, 264). This is a vivid example of an unfolding geography of epistemic power. We can observe the spread of a dominant

scientific practice into new political spaces defined by the discursive geographies of the object of mutual concern, i.e. climate change and its attendant geography of vulnerability, which brings with it an epistemic dependence on the assemblage of scientific and political instruments that configure our responses to the changing climate. Accounting for such performances of epistemic power means we must take concern with how science makes space for itself; with how it comes to define spaces of need or vulnerability, before rendering their extrication captive to its own epistemic practices.

PRECIS is a unique scientific tool that can undoubtedly generate great benefits for countries that use it in preparing climate change adaptation plans. Through the ability to visualise future climates, political bodies can prepare for coming changes and participate in international debates on a more equal footing with those countries who have traditionally dominated scientific discourse. But while PRECIS has the potential to empower, it also has the capacity to disempower, through the extension of a dominant epistemology that may freeze-out other ways of knowing the climate and of anticipating change. This is achieved through the establishment of relationships of dependency within knowledge-making networks which are in danger of reproducing historical gradients of power between political and epistemic centres and their peripheries. The tools of STS can help us unlock these complex geographies of epistemic power. If geographers engage seriously with this task, then debates about how we should best approach the challenges of a changing climate will be greatly enriched.

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Notes

- 1 Providing Regional Climates for Impacts Studies.
- 2 See the website: <http://www.facebook.com/precisrcm>.

- 3 The other three 'moments' are intersement – the processes by which allies are 'locked' into their roles within the wider work programme; enrolment – the definition and coordination of roles; and mobilisation – the verification that the spokespersons assigned to various collectivities are properly representative of their charges.
- 4 Global Climate Observing System.

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